YENEPOYA
Deemed to be University

Department of Statistics

Yenepoya (Deemed to be University)

Two Days Workshop on Programming in Python

Day 1: Foundation & Data Preparation			27.04.2026
Session No	**Session Title**	**Content Details**	**Facilitator**
1	**Introduction to Python**	• Environment setup (Anaconda, VS Code, Jupyter) • Syntax, indentation, and basic data types • Control Flow: If-else, loops, and list comprehensions • Defining functions and modular code	Dr. Rathnakar Shetty P
2	**Numerical Foundations with NumPy**	• Introduction to NumPy arrays and vectorization • Array operations and broadcasting • Mathematical and statistical functions in NumPy • Performance benefits over standard Python lists	Mr. Vasudev Shenoy
3	**Data Preparation with Pandas**	• Data structures: Series and DataFrames • Loading data from CSV, Excel, and SQL • Data Inspection: .head(), .info(), and .describe() • Indexing, slicing, and filtering datasets	Mr. Vasudev Shenoy
4	**ETL & Statistical Data Cleaning**	• Handling missing values and duplicates • Data type conversion and column renaming • GroupBy operations and pivot tables for summary statistics • Merging and joining disparate datasets	Mr. Vikyath

YENEPOYA
Deemed to be University

Department of Statistics

Yenepoya (Deemed to be University)

Two Days Workshop on Programming in Python

Day 2: Analytics, Visualization & Modeling			28.04.2026
Session No	**Session Title**	**Content Details**	**Facilitator**
5	**Analyzing & Visualizing Data with Matplotlib and Plotly**	• Introduction to Interactive Visualization • Creating Bar charts, Line graphs, and Scatter plots • Advanced visuals: Box plots and Histograms for distribution analysis • Customizing layouts, themes, and interactivity	Ms. Raheema
6	**Exploratory Data Analysis (EDA)**	• Identifying trends, outliers, and correlations • Multivariate analysis using Plotly Express • Using visualization to inform data cleaning decisions • Building a mini-dashboard within Jupyter Notebooks	Ms. Raheema
7	**Predictive Modeling: Fundamentals**	• Introduction to Scikit-Learn • Understanding supervised learning: Regression vs. Classification • Feature selection and splitting data (Train/Test split) • Linear Regression: Concepts and implementation	Mr. Vikyath
8	**Advanced Analysis & Model Evaluation**	• Logistic Regression for classification tasks • Model Evaluation: R-squared, MAE, and Confusion Matrix • Introduction to Random Forests or Decision Trees • Next steps: Deploying models and continuing the learning path	Mr. Vikyath